

WASP-12b

R-0559

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_170312 · created 2026-07-23T15:59:06+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457824.70961 +/- 0.00037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.119 +/- 0.038
Transit depth (Rp/Rs)^2	1.41 +/- 0.9 [%]
Orbital Inclination (inc)	79.36 +/- 2.84
Airmass coefficient 1 (a1)	1.239 +/- 0.05
Airmass coefficient 2 (a2)	-0.1256 +/- 0.0094
Scatter in the residuals of the lightcurve fit is	3.99 %
Best Comparison Star	None
Optimal Aperture	13.57
Optimal Annulus	33.16
Transit Duration (day)	0.112 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.119 $\pm$ 0.038 vs 0.1178 (deviation 0.03 σ)
Duration fit vs published	0.112 d vs 0.125 d (deviation 1.02 σ)
Mid-time O-C	-1.1 min
Depth SNR	2.9
Residual scatter	3.99 %

Light curve

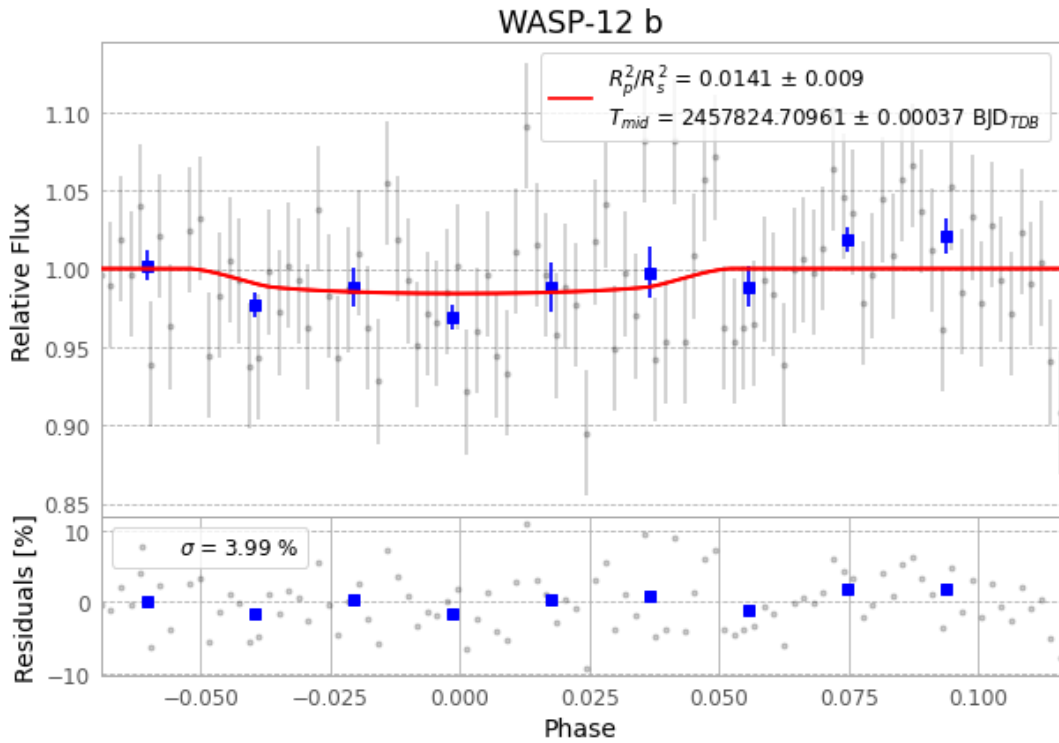


Figure 1 — FinalLightCurve_WASP-12 b_2017-03-11.png (SHA-256 42f78eb5134f04d5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [279, 363], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e5db50b5ef58846d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

98 FITS frames (64 MB), WASP-12170312030913.FITS → WASP-12170312080012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-170312.log (d94404312d72...), FinalLightCurve_WASP-12 b_2017-03-11.png (42f78eb5134f...), FinalLightCurve_WASP-12 b_2017-03-11.csv (33611594fd9d...)

Compute resources

Wall time 165.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-23 15:59Z.